

Phase 4 Report

CI/CD Pipeline, Monitoring & Cost Management

Student Study Planner — AWS Capstone Project

June 2026

Phase 4 Overview

Challenge: Automate deployments and set up observability within Free Tier limits.

Activities Summary

Activity	Status
Create GitHub Actions CI/CD workflow	✓ Done
Install Git on EC2 and clone repository	✓ Done
Set up SSH deploy script for EC2 + Nginx reload	✓ Done
Configure CloudWatch Alarms (CPU > 80%, ALB 5xx > 5%)	✓ Done
Install & configure CloudWatch Logs Agent (7-day retention)	✓ Done
Create AWS Budgets Free-Tier alert	✓ Done
Update Trello board	? Pending
Peer code review via GitHub Pull Requests	? Pending

Task 1 — GitHub Actions CI/CD Workflow

A GitHub Actions workflow was created at `.github/workflows/deploy.yml` to automatically deploy application code to the EC2 instance on every push to the main branch.

Workflow File (deploy.yml)

```
name: Deploy to EC2

on:
  push:
    branches:
      - main

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - name: SSH into EC2 and deploy
        uses: appleboy/ssh-action@v1.0.0
        with:
          host: ${ secrets.EC2_HOST }
          username: ${ secrets.EC2_USER }
          key: ${ secrets.EC2_SSH_KEY }
          script: |
            set -e
            cd /home/ec2-user/aws-capstone-project
            git pull origin main
            sudo cp -r app/* /var/www/study-planner/
            sudo chown -R nginx:nginx /var/www/study-planner
            sudo nginx -t
            sudo systemctl reload nginx
            echo "Deployment complete."
```

deploy.yml on GitHub

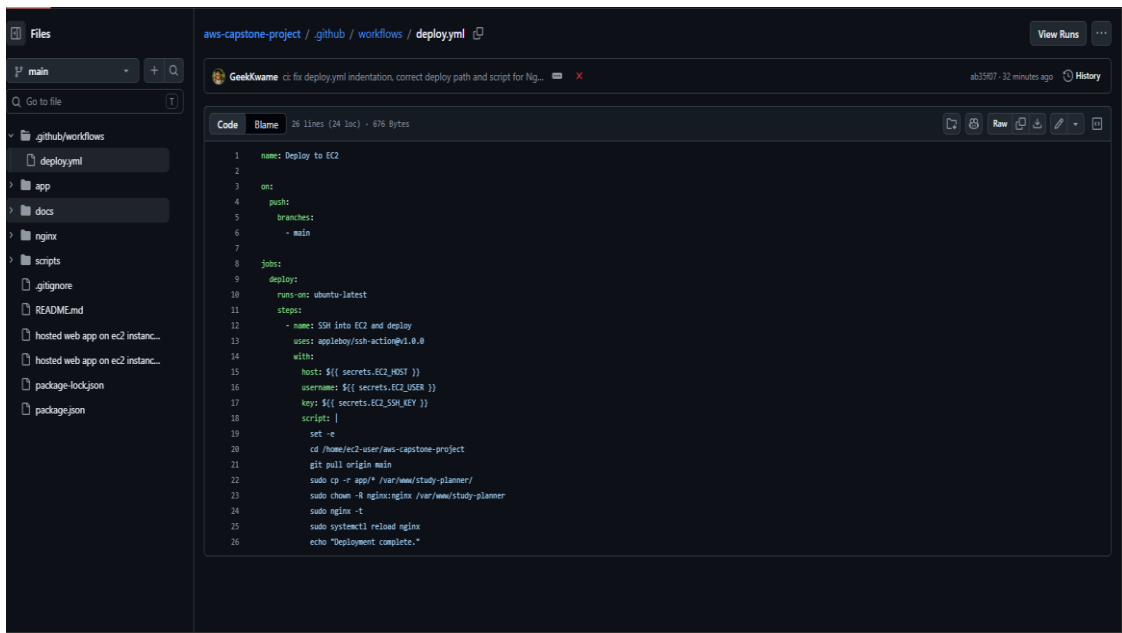


Figure: deploy.yml file as viewed on GitHub.

GitHub Actions Workflow Runs

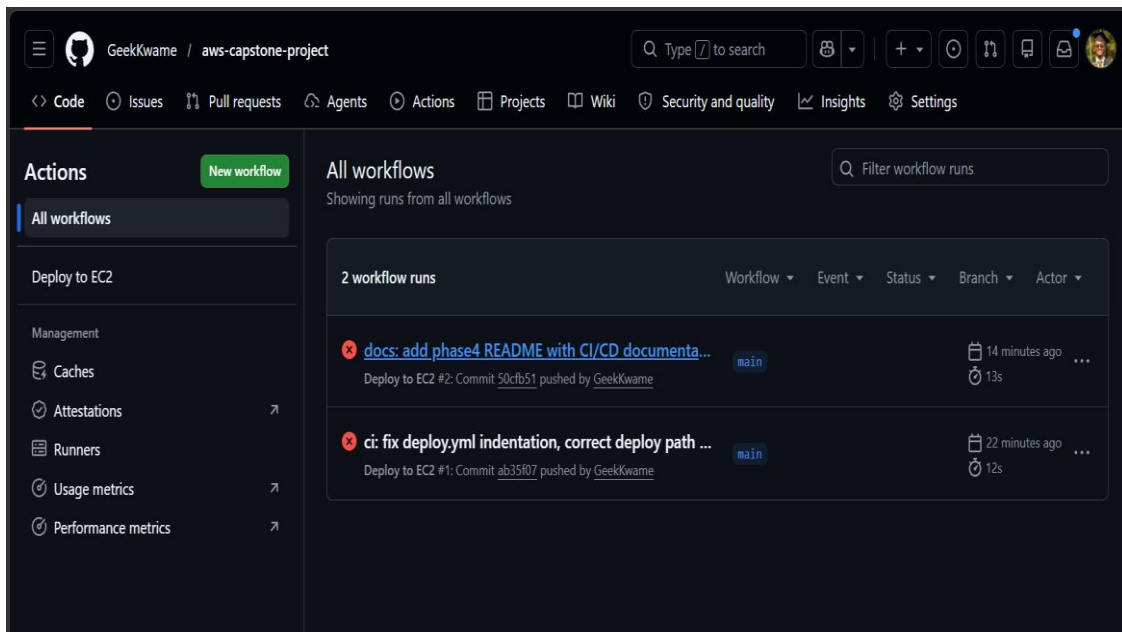


Figure: GitHub Actions tab showing the workflow.

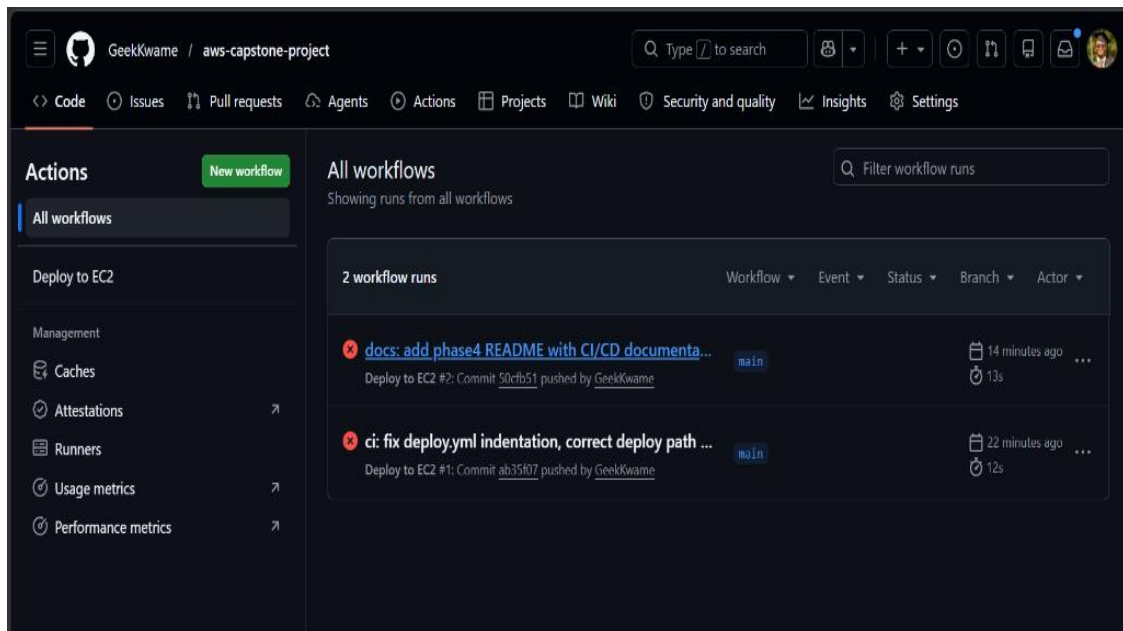


Figure: List of workflow runs triggered by pushes to main.

GitHub Repository Secrets

Three repository secrets are required for the workflow to SSH into EC2:

Secret Name	Description
EC2_HOST	EC2 Public IP — 3.237.34.20
EC2_USER	SSH username — ec2-user
EC2_SSH_KEY	Full contents of the SSH private key (.pem)

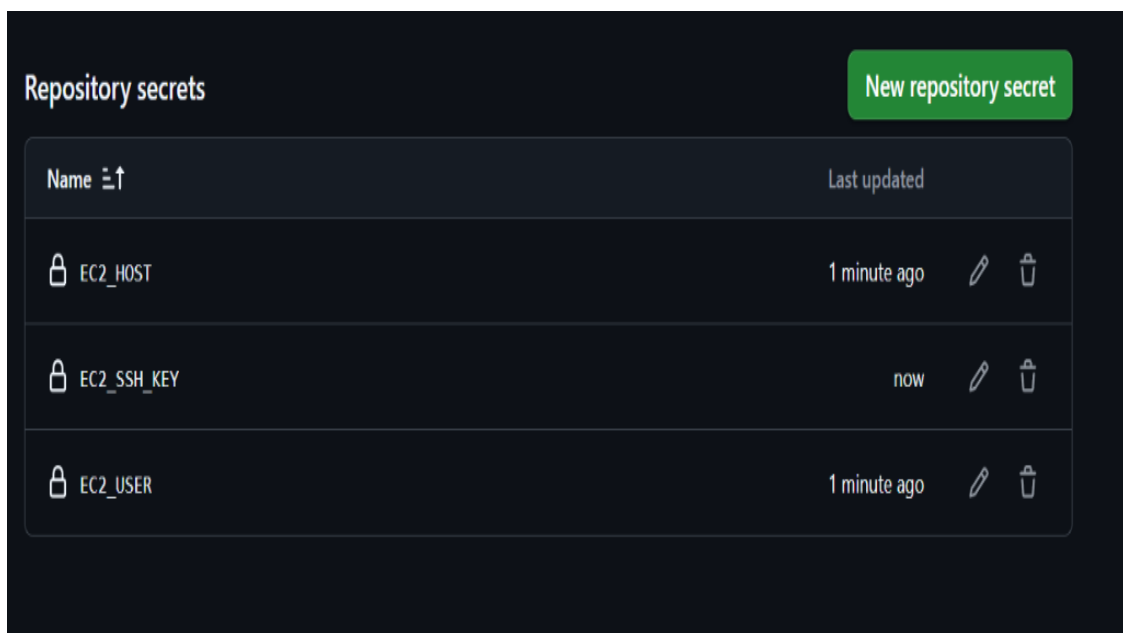


Figure: EC2_HOST, EC2_USER, and EC2_SSH_KEY configured as Actions secrets.

Bugs Fixed in Original deploy.yml

Bug	Fix Applied
on: and jobs: had 3 leading spaces — invalid YAML	Fixed to column-0 alignment
Deploy path 'your-app-folder' was a placeholder	Changed to /home/ec2-user/aws-capstone-project
npm install + pm2 restart — wrong for static Nginx site	Replaced with cp, chown, nginx -t, systemctl reload

Task 2 — Git Installation & Repository Clone on EC2

Git was not pre-installed on the Amazon Linux 2023 instance. It was installed and the GitHub repository was cloned to enable git pull during CI/CD deployments.

Commands Run on EC2

```
# Install Git
sudo dnf install -y git
git --version # git version 2.50.1

# Clone the repository
git clone https://github.com/GeekKwame/aws-capstone-project.git \
/home/ec2-user/aws-capstone-project
```

Verification

```
--- Repo app files ---
css  health.html  index.html  js

--- Nginx web root ---
css  health.html  index.html  js
```

Both directories match, confirming the deploy script will copy the correct files on future pushes.

Task 3 — CloudWatch Alarms

CloudWatch Alarms were set up to monitor infrastructure health and alert the team via email when the EC2 instance is under heavy load or the ALB starts returning HTTP 5xx errors.

1. EC2 CPU Utilization Alarm (EC2-High-CPU)

Property	Value
Metric	EC2 > Per-Instance Metrics > CPUUtilization
Threshold	Greater than 80%
Evaluation Period	3 consecutive periods of 5 minutes
SNS Topic	ec2-cpu-alert
Alarm Name	EC2-High-CPU

Configuration Steps

- Navigated to CloudWatch → Alarms → Create Alarm.
- Clicked Select metric → EC2 → Per-Instance Metrics → selected CPUUtilization.
- Set threshold type to Static, value to 80.
- Set evaluation periods to 3 out of 3.
- Created new SNS topic ec2-cpu-alert and subscribed the team email.
- Clicked Create alarm and confirmed the email subscription.

Screenshots

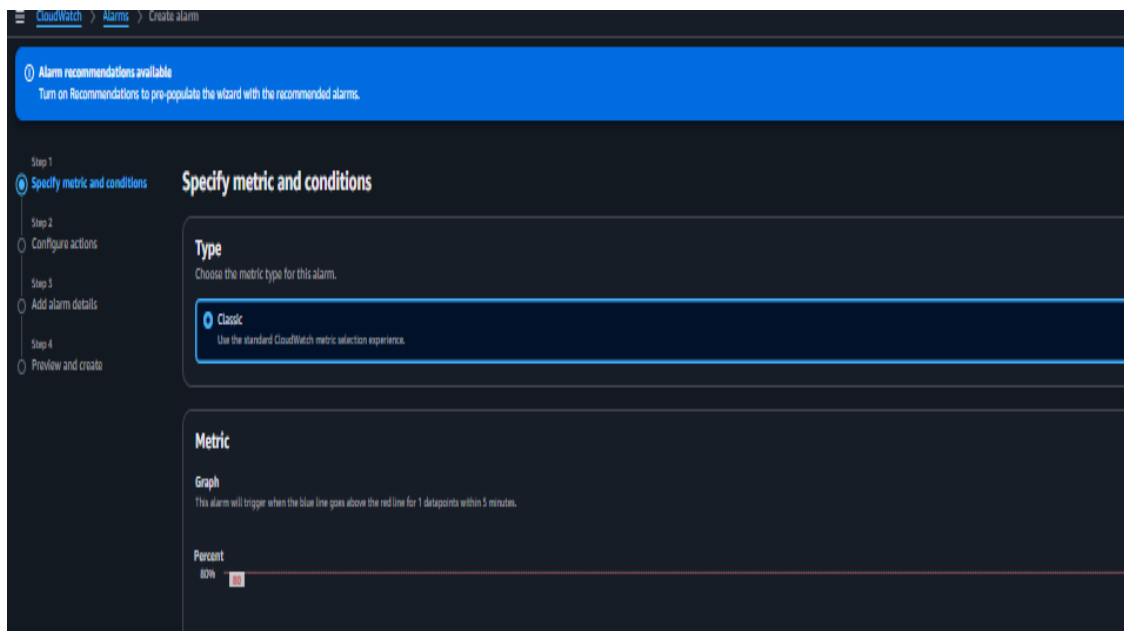


Figure: Selecting the CPUUtilization metric for the EC2 instance.

Alarm recommendations available
Turn on recommendations to pre-configure the alarm with the recommended alarm.

Specify metric and conditions

Type
☒ **Cases**
Use the standard CloudWatch metric selection experience.
 ☐ **Threshold**
Use the Threshold Guard to define the alarm.

Metric
 The alarm will trigger when the CloudWatch metric value for the specified metric is greater than the specified threshold for the specified period.

Graph
 The alarm will trigger when the CloudWatch metric value for the specified metric is greater than the specified threshold for the specified period.

Period
 5m

Threshold
 80%

Statistics
 Average

Dimensions
 InstanceId: i-1234567890

Conditions
 Threshold type: ☒ **State**
 When the metric value is greater than the specified threshold.

When the metric value is greater than the specified threshold
☒ **Greater/Equal**
☐ **Lesser/Equal**
☐ **Lesser**

Additional configuration

Cancel **Save**

Figure: Setting the alarm threshold to >80% for 3 consecutive 5-min periods.

Create alarm

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ **Create new topic**

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

user1@example.com, user2@example.com

Create topic

Add notification

Remove

Figure: Configuring SNS email notifications.

Figure: EC2-High-CPU alarm created and listed in the CloudWatch dashboard.

2. ALB 5xx Error Rate Alarm (ALB-High-5XX)

Property	Value
Metric	ApplicationELB > Per AppELB Metrics
Math Expression	$e1/m1 \times 100$ (HTTPCode_ELB_5XX_Count / RequestCount $\times$ 100)
Threshold	Greater than 5%
SNS Topic	ec2-cpu-alert (existing)
Alarm Name	ALB-High-5XX

Figure: Math expression alarm setup for calculating ALB 5xx error percentage.

Task 4 — CloudWatch Logs Agent

The Amazon CloudWatch Logs Agent was installed and configured on the EC2 instance to stream Nginx logs to AWS CloudWatch with a 7-day retention policy.

Installation

```
sudo yum install amazon-cloudwatch-agent -y
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\eddie\OneDrive\Desktop\aws-capstone-project> sudo yum install amazon-cloudwatch-agent -y
Sudo is disabled on this machine. To enable it, go to the Developer Settings page in the Settings app
PS C:\Users\eddie\OneDrive\Desktop\aws-capstone-project> ssh -i eddie-key.pem ec2-user@3.237.34.20
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Thu Jun 11 13:16:13 2026 from 154.161.129.223
[ec2-user@ip-172-31-71-213 ~]$ sudo yum install amazon-cloudwatch-agent -y
Last metadata expiration check: 1 day, 0:04:11 ago on Wed Jun 10 17:51:42 2026.
Dependencies resolved.

Package Arch Version Repository Size
Installing:
amazon-cloudwatch-agent
x86_64 1.300066.2-2.amzn2023 amazonlinux 68 M

Transaction Summary
Install 1 Package

Total download size: 68 M
Installed size: 231 M
Downloading Packages:
amazon-cloudwatch-agent-1.300066.2-2.a 65 MB/s | 68 MB 00:01
-----
Total 62 MB/s | 68 MB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
```

Figure: CloudWatch Agent installed on EC2 (version 1.300066.2-2).

Configuration File

Config file written to: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json

```
{
  "logs": {
    "logs_collected": {
      "files": {
        "collect_list": [
          {
            "file_path": "/var/log/nginx/access.log",
            "log_group_name": "study-planner-nginx",
            "log_stream_name": "{instance_id}",
            "retention_in_days": 7,
          },
          {
            "file_path": "/var/log/nginx/error.log",
            "log_group_name": "study-planner-nginx-errors",
            "log_stream_name": "{instance_id}",
            "retention_in_days": 7
          }
        ]
      }
    }
  }
}
```

```
GNU nano 8.3 /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json Modified
{
  "logs": {
    "logs_collected": {
      "files": {
        "collect_list": [
          {
            "file_path": "/home/ec2-user/your-app-folder/app.log",
            "log_group_name": "my-web-app",
            "log_stream_name": "{instance_id}",
            "retention_in_days": 7
          }
        ]
      }
    }
  }
}
```

Figure: Agent JSON config open in nano on EC2 — log paths, group, stream, and 7-day retention.

Starting the Agent

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config -m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -s
```

```
[ec2-user@ip-172-31-71-213 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config -m ec2 \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -s
***** processing amazon-cloudwatch-agent *****
Starting config-downloader, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments: [config-downloader -mode ec2 -download-source file:/opt/aw
s/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -output-dir /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d -config /op
t/aws/amazon-cloudwatch-agent/etc/common-config.toml -multi-config default]I! Trying to detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
Start configuration validation...
2026/06/11 17:58:45 Reading json config file path: /opt/aws/amazon-cloudwatch-agent.d/file_amazon-cloudwatch-agent.js
on.tmp ...
2026/06/11 17:58:45 I! Valid Json input schema.
2026/06/11 17:58:45 Configuration validation first phase succeeded
Starting config-translator, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments: [config-translator -output /opt/aws/amazon-cloudwatch-agen
t/etc/amazon-cloudwatch-agent.toml -mode ec2 -config /opt/aws/amazon-cloudwatch-agent/etc/common-config.toml -multi-config default -input /opt/aw
s/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json -input-dir /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d]I! Trying to
detect region from ec2
D! [EC2] Found active network interface
I! imds retry client will retry 1 times
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.tom
l
Configuration validation second phase succeeded
Configuration validation succeeded
amazon-cloudwatch-agent has already been stopped
Created symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/systemd/system/amazon-cloudwatch-agent.service
.
[ec2-user@ip-172-31-71-213 ~]$
```

Figure: Configuration validation succeeded and CloudWatch Agent service started.

Summary

Property	Value
Log Group (access)	study-planner-nginx
Log Group (errors)	study-planner-nginx-errors
Log Stream	{instance_id} (auto-resolved per instance)

Retention	7 days
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Task 5 — AWS Budgets

An AWS Budget alert was created to monitor monthly spending and send an email notification the moment any cost above \$0.01 is incurred, protecting the team from unexpected Free Tier overages.

Configuration Steps

- Navigated to AWS Console → Billing → Budgets → Create budget.
- Selected the Zero spend budget template (alerts on any spend above \$0.01).
- Named the budget: Free-Tier-Monitor.
- Added the team lead email address as a notification recipient.
- Clicked Create budget.

The screenshot shows the 'Create budget' wizard in the AWS Billing console. At the top, there's a section for 'Billing View - optional' with a note about data export. Below this is the 'Templates - new' section, which prompts the user to 'Choose a template that best matches your use case.' There are four radio button options: 'Zero spend budget' (selected), 'Monthly cost budget', 'Daily Savings Plans coverage budget', and 'Daily reservation utilization budget'. Each option has a brief description. Below the templates is the 'Zero spend budget - Template' section. It includes a 'Budget name' field with the value 'Free-Tier-Monitor', an 'Email recipients' field with a placeholder 'Separate email addresses using commas', and a 'Scope' section with a radio button option 'You will be notified via email when any spend above \$0.01 is incurred.' At the bottom, there's a 'Template settings' section with a note about default configurations and a 'Custom' link. The 'Create budget' button is highlighted in orange at the bottom right.

Figure: Zero spend budget template selected and named 'Free-Tier-Monitor'.

The screenshot shows the AWS Budgets console. At the top, a green banner indicates 'Your budget Free-Tier-Monitor has been created successfully.' Below this is the 'Budgets (1)' section with a search bar and a 'Type - Show all budgets' dropdown. To the right are buttons for 'Download CSV', 'Actions', and 'Create budget'. Below this is a table with columns: Name, Thresholds, Health status, Billing View, Budget, Amount used, Forecasted a..., and Current vs. budgeted. The table contains one row for the 'Free-Tier-Monitor' budget, which is in a 'Healthy' state. The budget amount is \$1.00, and the current usage is 0.00%.

Name	Thresholds	Health status	Billing View	Budget	Amount used	Forecasted a...	Current vs. budgeted
Free-Tier-Monitor	OK	Healthy	Primary View	\$1.00	-	-	0.00%

Figure: Free-Tier-Monitor budget active — Status: Healthy, 0.00% used.

Budget Summary

Property	Value
Budget Name	Free-Tier-Monitor
Budget Type	Zero spend budget
Alert Threshold	Any spend > \$0.01
Notification	Email to team lead
Status	Active & Healthy

Last updated: June 11, 2026 — Phase 4: CI/CD pipeline live, CloudWatch Alarms & Logs configured, AWS Budget active.